

SYNOPSIS

On 8 August 2020, an unnamed personal watercraft and the rigid inflatable boat *Rib Tickler* collided at high speed in the Menai Strait, Wales. A passenger on board *Rib Tickler* was struck by the personal watercraft and fatally injured.

The personal watercraft had been jumping across *Rib Tickler*'s wake when the rigid inflatable boat altered course across the personal watercraft's path. The personal watercraft collided with *Rib Tickler* and struck one of the passengers, who was seated on the aft bench seat.

The accident happened because the personal watercraft was too close to *Rib Tickler* when jumping its wake, which left insufficient time to react when the rigid inflatable boat's driver altered course across its path. *Rib Tickler*'s driver had commenced his turn without sufficiently checking astern for other craft. The investigation identified that neither *Rib Tickler*'s driver, its owner, nor the personal watercraft rider had attended an appropriate training course and the knowledge and skill levels of the persons in control or overseeing the two craft were not appropriate for the manoeuvres being undertaken. The investigation concluded that the rigid inflatable boat's driver did not have full awareness of other water users before commencing his turn and the personal watercraft rider did not have sufficient knowledge or experience to appreciate the hazards involved in wake jumping.

The investigation also identified that although the Isle of Anglesey County Council had a marine safety management system there was insufficient local governance of the marine environment, with a lack of resources, risk assessment for leisure users, or powers to manage their waterspace. Improvement areas for Royal Yachting Association training and guidance have also been highlighted in light of the safety issues raised. The report also explored the legislative status of personal watercraft in the United Kingdom.

Following this accident, Isle of Anglesey County Council has engaged a maritime specialist to conduct a review into its governance of the Menai Strait, and the Royal Yachting Association has updated its course content and guidance documents.

Recommendations have been made to the Isle of Anglesey County Council about its management of the Menai Strait waterspace, reviewing its legislation governing the Menai Strait and resource availability within its maritime team. A recommendation has also been made to the Personal Watercraft Partnership and Royal Yachting Association to formalise the creation of a cross-industry group, focusing on a consistent nationwide approach to personal watercraft management.

SECTION 1 – FACTUAL INFORMATION

1.1 PARTICULARS OF *RIB TICKLER*, THE PERSONAL WATERCRAFT, AND ACCIDENT

SHIP PARTICULARS		
Vessel's name	<i>Rib Tickler</i>	Not applicable
Flag	United Kingdom	Not applicable
Type	Ribeye S650	Yamaha WaveRunner VX Cruiser
Registered owner	Private	Private
Construction	Glass-Reinforced Plastic hull, Hypalon tubes	Glass-Reinforced Plastic hull
Year of build	2008	2011
Length overall	6.5m	3.27m
VOYAGE PARTICULARS		
Port of departure	Menai Bridge	Menai Bridge
Port of arrival	Menai Bridge	Menai Bridge
Type of voyage	Leisure	Leisure
MARINE CASUALTY INFORMATION		
Date and time	8 August 2020 at 1928	
Type of marine casualty or incident	Very Serious Marine Casualty	
Location of incident	Menai Strait	
Place on board	Starboard aft	Bow
Injuries/fatalities	1 fatality	None
Damage/environmental impact	Superficial damage to boat	Superficial damage to personal watercraft
Ship operation	On passage	On passage
Voyage segment	Mid-water	Mid-water
External & internal environment	Northerly wind 5 knots, smooth water, good visibility	
Persons on board	4	1

1.2 BACKGROUND

Rib Tickler was a privately-owned 6.5m rigid inflatable boat (RIB). Its owner and his family lived in the Midlands and kept a caravan near the Isle of Anglesey (**Figure 1**), which they used most weekends in the summer. The family also owned two personal watercraft¹ (PW). During his visits to Anglesey, *Rib Tickler*'s owner often met up with a local friend, who was the RIB driver² at the time of the accident, and the friend's brother, who were both PW owners, and their families regularly spent time together.

¹ A personal watercraft refers to a watercraft that the rider sits or stands on. The term Jet Ski is the trademarked brand of personal watercraft manufactured by Kawasaki. Other popular types in the British market are the Yamaha WaveRunner and the Bombardier Sea-Doo.

² For clarity, the RIB driver at the time of the accident will be referred to as the RIB driver throughout the report.

1.3 NARRATIVE

On 7 August 2020, *Rib Tickler*'s owner travelled to Anglesey with his mother, Jane Walker, and his father for a weekend at their caravan. The following day (8 August) the weather was good, so *Rib Tickler*'s owner contacted his friend, the RIB driver, and arranged to meet him and his family later that afternoon at Menai Bridge to go out onto the water.

At about 1630, *Rib Tickler*'s owner and his parents arrived at the Porth y Wrach slipway (**Figure 2**) and launched the RIB. At about 1730, the RIB driver arrived at the slipway with his family and two PW. During the next hour, members of the RIB driver's family took turns riding the PW and occasionally rafted alongside *Rib Tickler* to share drinks and chat with the RIB owner and his parents. The RIB driver and the rest of his family sat on the slipway, enjoying the fine weather.

Image courtesy of Google Earth (www.google.co.uk/earth)



Figure 2: Google Earth image of the Menai Bridge, with inset showing Porth y Wrach slipway

At about 1830, the RIB driver joined *Rib Tickler*'s owner on board the RIB. They then left the Menai Bridge area through the Swellies³ with the PW, which were being ridden by one of the owner's sons and his niece. The PW riders carried out various high-speed manoeuvres, such as wake jumping, as the group made their way to the Plas Newydd National Trust property (**Figure 3**) where they stopped and rafted together. After a short while, the PW riders set off towards the Britannia Bridge (**Figure 4**) with *Rib Tickler* following on behind. During the transit back through the Swellies, the PW riders dropped back behind *Rib Tickler* and started to zig-zag and jump across its wake.

Reproduced from Admiralty Chart BA 1464 by permission of HMSO and the UK Hydrographic Office

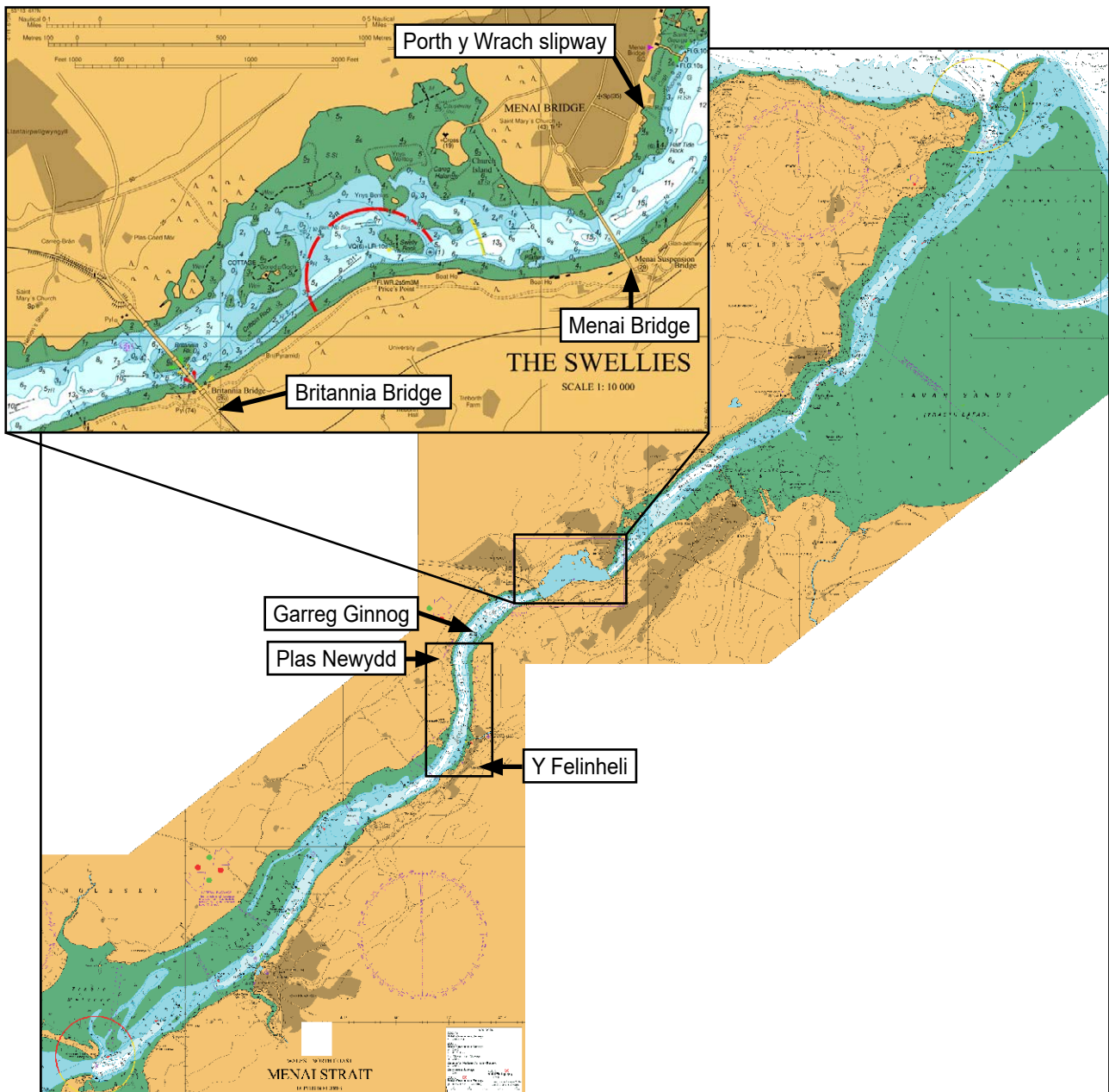


Figure 3: British Admiralty Chart 1464, showing the Menai Strait

³ The stretch of water on the Menai Strait between the Menai Bridge and the Britannia Bridge.

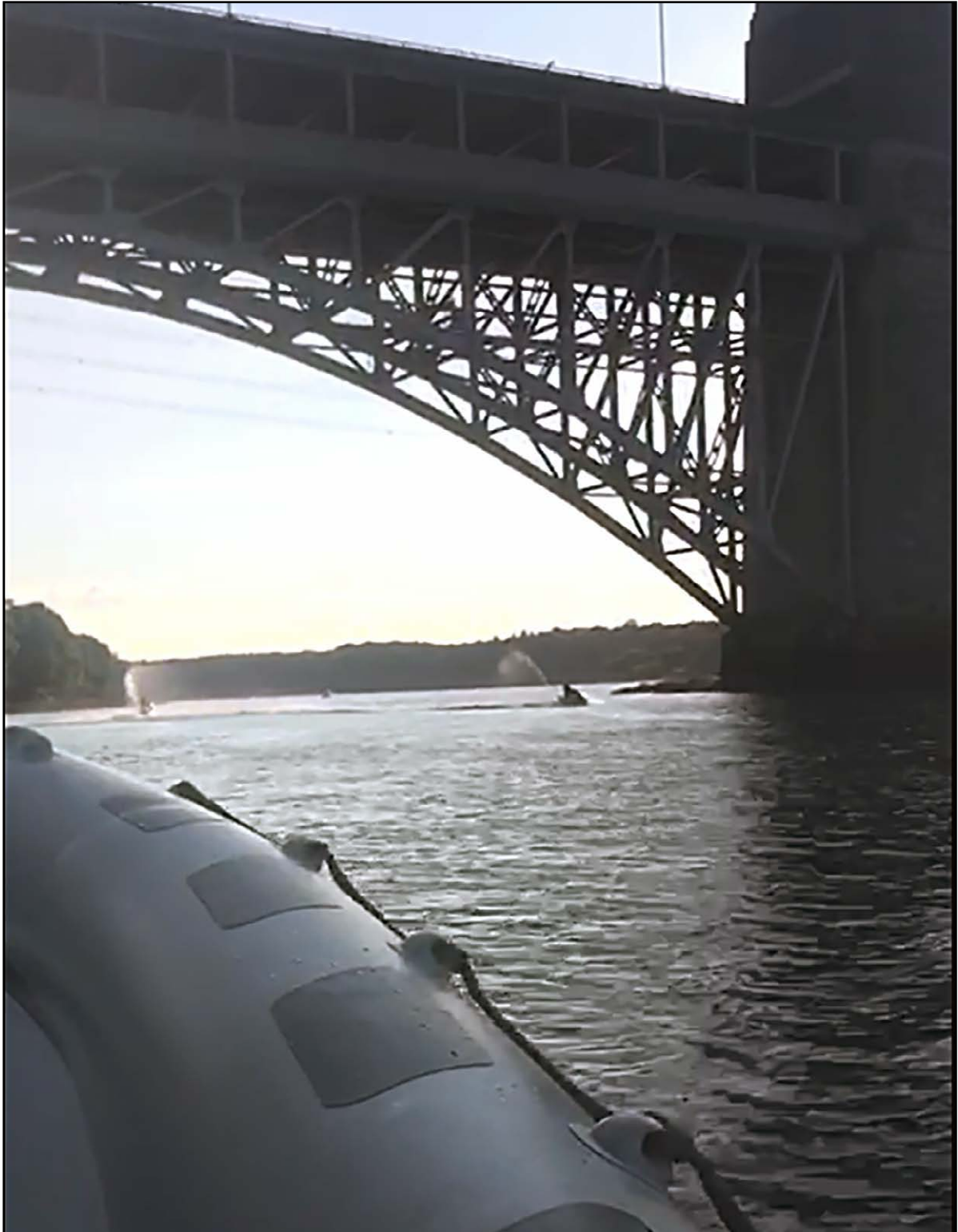


Figure 4: Video still taken from *Rib Tickler* of the group approaching Britannia Bridge

When they reached Porth y Wrach, *Rib Tickler*'s owner asked his friend if he wanted to drive the RIB. His friend accepted the offer, attached the engine kill cord⁴ and took the helm. *Rib Tickler*'s owner was standing on the port side of the RIB's centre console as his friend set the RIB on a north-easterly direction towards Bangor. The owner's father was sitting on the port side of the RIB's aft bench seat and his mother was sitting on the starboard side of the bench seat, facing slightly to port (**Figure 5**).

⁴ A safety device intended to stop the engine if the helm is thrown overboard or away from the helm.

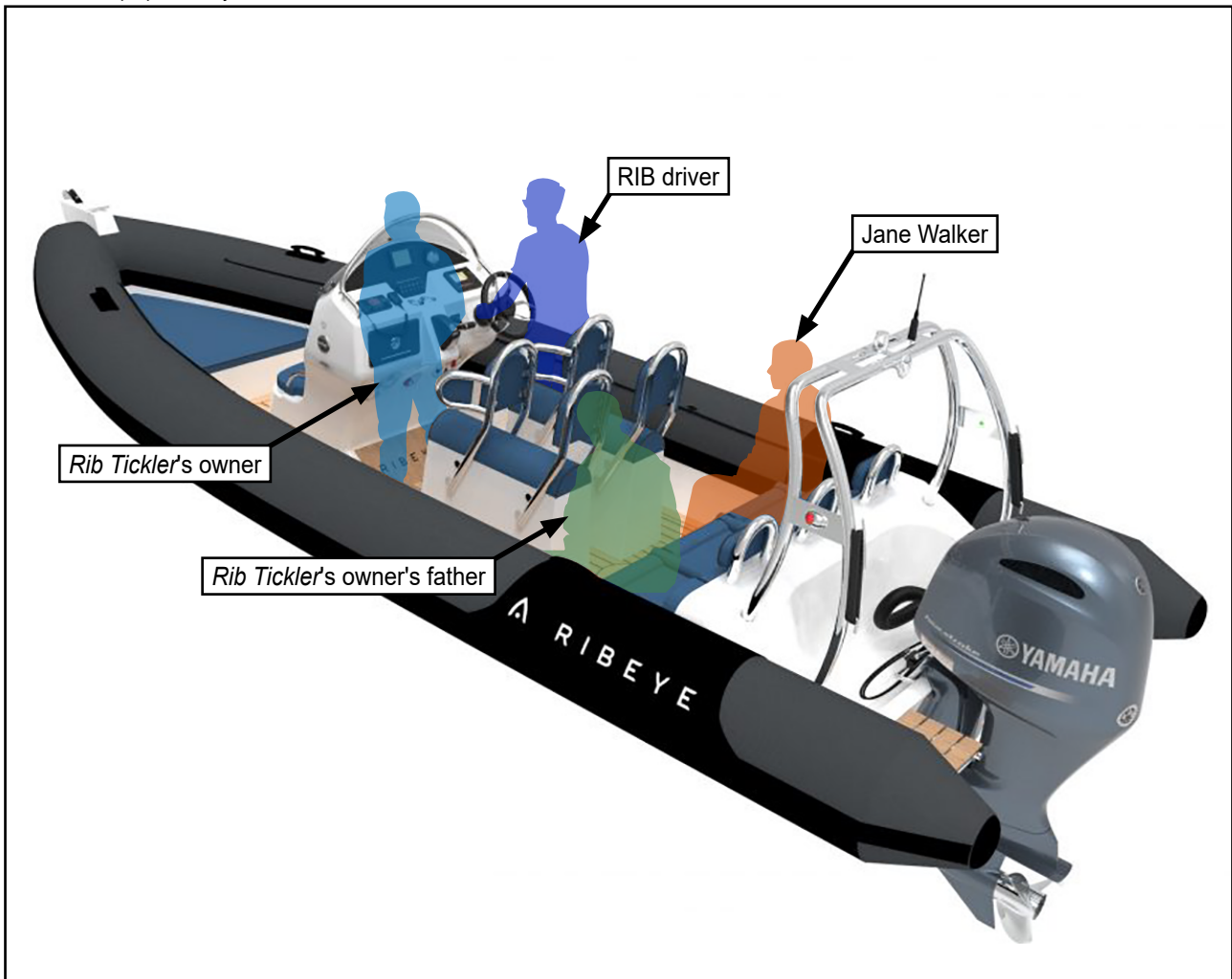


Figure 5: Ribeye S650, showing seating positions on *Rib Tickler* at the time of the accident

While steering *Rib Tickler* on a straight course, the RIB driver began to increase speed, and within a couple of minutes the RIB was close to full throttle (**Figure 6**) and travelling at about 25 to 30 knots (kts).

At 1928, a member of the RIB driver's family, who was following behind *Rib Tickler* on one of the PWs, positioned herself for a high-speed diagonal run from left to right across the RIB's wake. As she accelerated forward and hit the wake, the RIB driver looked over his right shoulder and believing it clear made a sharp turn to starboard. *Rib Tickler's* owner, who had also looked over his right shoulder, saw that the RIB was cutting across the path of the PW and shouted a warning (**Figure 7**).

The PW rider was unable to take avoiding action and the PW and *Rib Tickler* collided. The PW hit the RIB owner's mother and knocked her sideways from the starboard side of the aft bench seat and onto her husband (**Figure 8**). The PW also made contact with the RIB's aft A-frame before pivoting and sliding down the starboard side, landing upright in the water with its rider still on board.

Following the collision, *Rib Tickler's* owner took immediate control of the helm and headed towards the slipway. His mother was lying unconscious on the bench seat with her husband attending to her; the RIB driver called 999 and asked for the emergency services. As *Rib Tickler* approached the slipway, the RIB owner's mother regained consciousness.

While waiting for the emergency services to arrive, the group recovered the RIB onto its trailer at the bottom of the slipway (**Figure 2**). Within 12 minutes, two North Wales ambulance service paramedics had arrived on scene and began assessing and then treating the casualty on board the RIB. At 2008, an ambulance arrived and the RIB owners' mother, who had again become unconscious, was transferred from the RIB onto a trolley and then into the ambulance. She was declared deceased on arrival at hospital.

The RIB driver, who was helming the RIB at the time of the accident, and the PW rider were breathalysed following the accident and found to be under the alcohol limit for driving on the road⁵.

A postmortem examination identified that the casualty's death was caused by head, neck, and chest injuries, which were detailed in the pathologist's report as *essentially unsurvivable*. The RIB owner's father also suffered significant bruising to his right shoulder area.

Rib Tickler suffered minor impact damage to the starboard side bench seat head rest and the starboard side A-Frame stanchions. The starboard navigation light was also broken, and there were scuff marks on the starboard tube (**Figure 9**). The PW suffered minor damage to the gel coat and the fender strip on the port side (**Figure 10**).

1.4 ENVIRONMENTAL CONDITIONS

The wind was a Beaufort force two from the north, which resulted in calm waters in the Menai Strait. The temperature was 18°C with cloudless skies. At the time of the accident, the tidal stream was running north-east at a rate of 1kt.

1.5 QUALIFICATIONS AND EXPERIENCE OF THOSE INVOLVED

Rib Tickler's owner had completed the Royal Yachting Association (RYA) Personal Watercraft Proficiency course, which was a local requirement for riding PW on a lake he had previously used. He had ridden PW on the Menai Strait over a period of about seven years and *Rib Tickler* during 2019. He did not hold any powerboat qualifications.

The RIB driver had ridden his PW on the Menai Strait for about five years. He had not attended the RYA Personal Watercraft Proficiency course and held no powerboat qualifications. He had helmed a RIB once before, about 25 years earlier.

The PW rider had driven a PW eight or nine times prior to the accident. She used her father's PW and had only ridden in company when her family went to the Menai Strait. She held no PW or powerboat qualifications. At the time of the accident, she was 17 years old.

All members of both families were wearing buoyancy aids while on the water, and the PW riders had handheld very high frequency (VHF) radios attached to their buoyancy aids. They commonly jumped each other's wakes when on the water together.

⁵ In England, Wales and Northern Ireland, the alcohol limit for drivers is 80 milligrams of alcohol per 100 millilitres of blood, 35 micrograms per 100 millilitres of breath (<https://www.gov.uk/drink-drive-limit>).

1.7 THE PERSONAL WATERCRAFT

The PW was a 2011 model Yamaha WaveRunner VX Cruiser. It was powered by a 1052 cubic capacity, four-cylinder engine, had a maximum capacity of three people and a maximum speed of approximately 56kts.

A manufacturer's warning label (**Figure 11**) was attached to the PW's engine cover that provided guidance on how to avoid the risk of serious injury or death. The warnings on the label included:

RIDE WITHIN YOUR LIMITS AND AVOID AGGRESSIVE MANEUVERS to reduce the risk of loss of control, injuries, and collisions.

This is a high performance boat – not a toy. Sharp turns or jumping wakes or waves can increase the risk of back/spinal injury (paralysis), facial injuries and broken legs, ankles, and other bones. Do not jump wakes or waves.

*Collisions result in more **INJURIES AND DEATHS** than any other type of accident for personal watercraft (PWC).*

TO AVOID COLLISIONS:

SCAN CONSTANTLY for people, objects, and other watercraft. Be alert for conditions that limit your visibility or block your vision of others.

OPERATE DEFENSIVELY at safe speeds and keep a safe distance from people, objects, and other watercraft.

- *Do not follow directly behind PWCs or other boats.*
- *Do not go near others to spray or splash them with water.*
- *Avoid sharp turns or other maneuvers that make it hard for others to avoid you or understand where you are going.*
- *Avoid areas with submerged objects or shallow water.*

TAKE EARLY ACTION to avoid collisions. Remember, PWCs and other boats do not have brakes. [sic]

1.8 MENAI STRAIT

The Menai Strait is a narrow stretch of tidal water that separates the Isle of Anglesey from the mainland of Wales. It varies in width between 8000m at its north-eastern end to less than 400m at its narrowest point.

The town of Menai Bridge is situated at the termination of the Menai Suspension Bridge on the Isle of Anglesey. The Porth y Wrach slipway is open to the public and gives access to swing moorings that extend, almost unbroken, north-east to Beaumaris (**Figure 3**).

The tidal stream in the Menai Strait can be significant, especially in an area known as the Swellies, which lies between the Britannia Bridge and the Menai Suspension Bridge where, on spring tides, the current can reach 8kts.

1.9.3 Marine safety management system

The PMSC requires that an MSMS that manages the port's hazards and risks be developed, implemented, and maintained.

Two important aspects of any MSMS include:

- The undertaking of a navigation risk assessment considering all foreseeable activities within the port or harbour area, and
- the ability of the harbour authority to record, investigate and analyse incidents in order to determine underlying marine safety deficiencies. Once identified, corrective action can then be taken to reduce the risk to a level as low as reasonably practicable.

The Maritime and Coastguard Agency (MCA) can conduct health checks to measure a port's compliance with the PMSC and, where appropriate, identify ways to enhance it. Health checks also aim to identify and share best practice. A health check visit is intelligence-led and may be triggered for a variety of reasons; the port or facility may also volunteer for one.

1.9.4 Menai Strait governance

The responsibility for safe navigation in the Menai Strait is shared between the Caernarfon Harbour Trust (CHT) and Isle of Anglesey County Council (IACC).

CHT is the SHA responsible for the regulation and provision of safe navigation between Caernarfon Bay and the Menai Suspension Bridge (**Figure 3**). CHT is also the Competent Harbour Authority, responsible for providing pilotage for the entire Menai Strait.

IACC is the SHA for the section of water between the Menai Suspension Bridge and Ynys Faelog (**Figure 2**), which includes the harbour of Menai Bridge and St. George's Pier.

Other interested parties include Gwynedd Council, who hold responsibility for several launching sites that exist on the southern bank of the Menai Strait, and Conwy County Borough Council, who are the SHA for the port of Conwy, at the north-eastern extreme of the Menai Strait.

IACC had a speed limit of 8kts in bathing areas and advised the same limit within moorings. CHT had three 5kts speed limit zones in place, including at an area known as Y Felinheli (**Figure 12**). As an immediate reaction to the accident at Menai Bridge, CHT extended the 5kts speed limit further up the Menai Strait towards the Britannia Bridge and past the Plas Newydd National Trust property to Garreg Ginnog (**Figure 3**).

1.10 ISLE OF ANGLESEY COUNTY COUNCIL

1.10.1 Maritime responsibilities

IACC's executive committee were individually and collectively responsible for ensuring that PMSC requirements were met. IACC employed an independent consultant as their DP responsible for assessing and reporting on the effectiveness of the MSMS in complying with the PMSC.

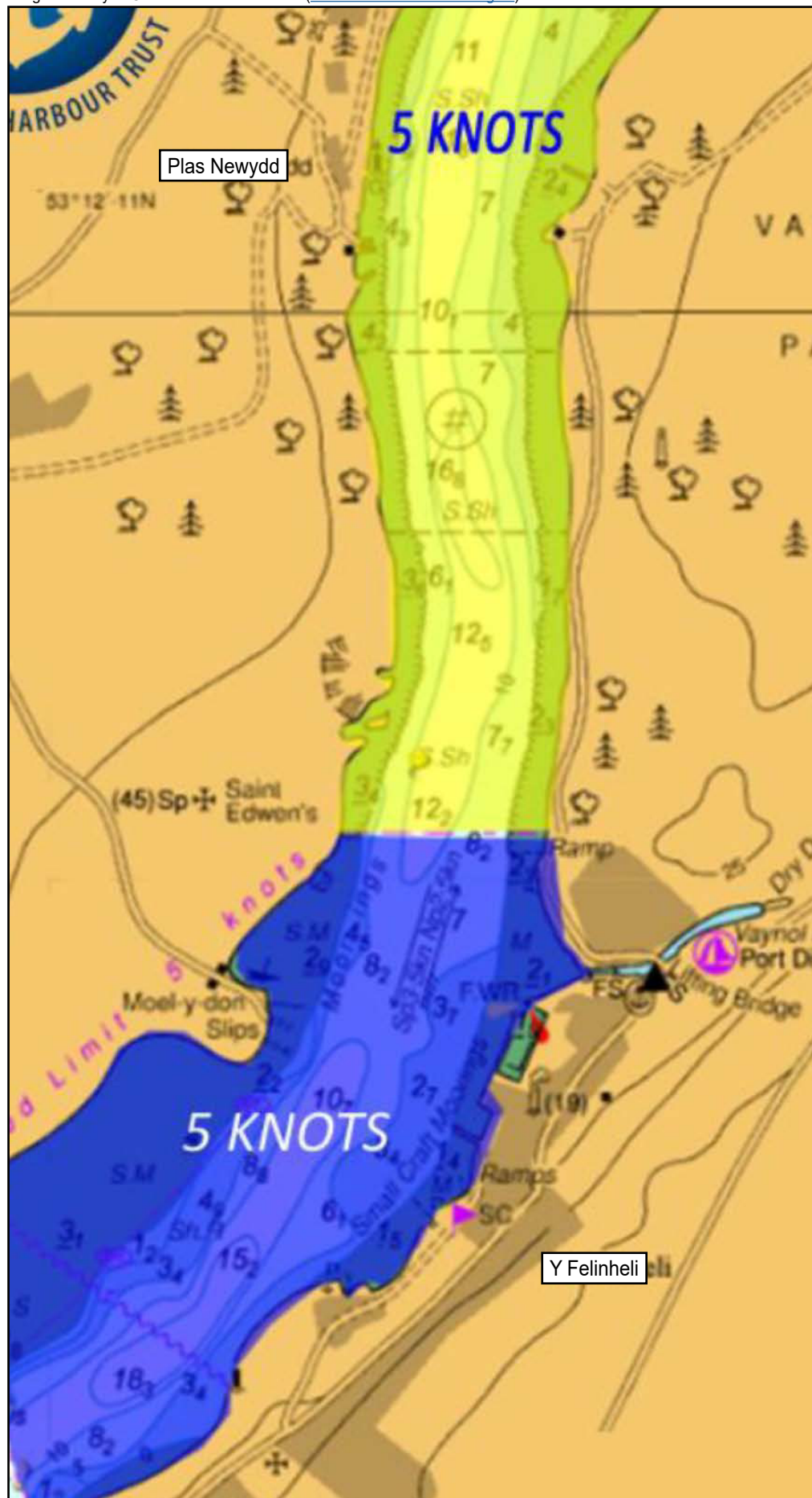


Figure 12: Caernarfon Harbour Trust extended speed limit

Within IACC a senior maritime officer managed all marine-related affairs around the island and was operationally responsible for MSMS implementation. IACC employed a harbourmaster to oversee the port of Amlwch; the ports of Menai Bridge and Beaumaris were overseen by a pier master who had day-to-day operational responsibility.

1.10.2 Current byelaws and harbour directions

The IACC document *Seashore and Seaside Pleasure Boats Byelaws* identified 26 areas around the Isle of Anglesey that had byelaws applicable to them. The document was last updated in 1996.

The only area in the Menai Strait covered by byelaws was the bathing area at Beaumaris, where there was a speed limit of 8kts between the months of April and September and the hours of 1000 and 1900. The byelaws stated that:

No person being the navigator of a pleasure boat shall cause or suffer such vessel to be driven or sailed in a dangerous manner or without due care and attention or without reasonable consideration for other persons.

There were no byelaws applicable to Menai Bridge, although there was an advisory speed restriction of 8kts within the moorings. There were no general, special, or harbour directions in force.

Both families were aware of the local 8kts advisory speed restriction within the moorings but, on the evening of the accident, one of the RIB driver's sons was challenged by an employee of a local commercial RIB ride company due to his excessive speed through the moorings.

1.10.3 Resources

IACC had one patrol boat, which was kept at St George's Pier (**Figure 2**) and used between Menai Bridge and Beaumaris as and when it was deemed necessary and staff availability allowed. The patrol boat provided an on-water presence and had been taken onto the water prior to the accident by the Senior Maritime Officer, but it was not equipped or intended to be used to chase PW or high-speed RIBs. Previously the council had owned three PW, but these were disposed of in 2014 due to budget pressures.

IACC employed a team of seasonal beach wardens and slipway attendants who provided a presence at beaches and slipways around the Isle of Anglesey. They wore branded clothing and identification badges, so were immediately recognisable. They were also first aid trained. Their primary task was health and safety, but they also provided local tourism information.

The slipway attendants checked the registration and insurance of those launching from the slipways and took registration and launching fee payments as necessary. The number of wardens and attendants employed had reduced from 26 in 2014 to 11 in 2020.

The recognised best practice for an over-the-shoulder pre-manoeuve check is to look over both shoulders to a line right astern of you, but this procedure is not explicitly explained in either the syllabus or handbook.

1.12.3 Powerboating scheme

The RYA offers six separate powerboating courses. At entry level are the Level 1 – Start Powerboating and Level 2 – Powerboat Handling courses. The Level 1 course is intended for participants to gain basic skills while the Level 2 course provides the skills and background knowledge that a competent boat driver needs.

The RYA's *Start Powerboating* publication contains guidance based on the curriculums of these two courses. Participants are required to understand the application of the International Regulations for Preventing Collisions at Sea, 1972 (COLREGs), especially Rule 5 and Rule 6, which govern lookout and safe speed respectively, as well as to have an awareness of other water users.

The publication's *Operating at Planing Speed* section also contains this advice:

When intending to travel at high speeds, you must keep a really good all-round lookout. Check it is clear to turn and be aware of other water users.

Similar to the guidance given in the PW handbook, there is no explicit instruction on how to apply best practice over-the-shoulder checks.

The practice of bringing two boats alongside each other while making way, known as Pacing⁹, is not covered in the RYA level 1 and 2 courses. Pacing used to be included in the RYA Level 2 syllabus, but it was removed due to concerns over the safety of the manoeuvre. As a result, all advice on operating in groups of craft was also removed. There is also no advice on the oversight of an inexperienced or untrained helm in an informal setting.

1.13 SIMILAR ACCIDENTS

1.13.1 Two fire and rescue service boats – collision

Two fire and rescue service boats were in a collision while undertaking boat training and familiarisation, which resulted in one fatality (MAIB report 17/2020¹⁰). The MAIB's investigation found that the collision occurred because both boats were operating at speed and carrying out uncoordinated manoeuvres in the same stretch of river.

As his boat exited a large turn made at speed, one helmsman attempted to pass between the other boat and the shoreline. The other boat's helmsman turned his craft sharply to port as part of his training manoeuvres, which resulted in the boats heading towards each other. The subsequent action taken by both boats to avoid a collision was unsuccessful. There was a lack of awareness by either helm of where the other boat was. The investigation found that the accident could have been avoided had the training been properly planned.

⁹ Pacing utilises the natural pressure zones created by the boat's movement through the water. This is useful for at sea transfers and is used primarily by maritime security forces and rescue organisations such as the Royal National Lifeboat Institution but is also applicable when training those in other craft.

¹⁰ <https://www.gov.uk/maib-reports/collision-between-2-fire-and-rescue-service-boats-with-loss-of-1-life>.

1.13.2 *Osprey* and *Osprey II* – collision

Two passenger carrying RIBs, *Osprey* and *Osprey II*, collided in the Firth of Forth. A passenger sitting on an inflatable tube on *Osprey II* was crushed between *Osprey*'s bow and *Osprey II*'s helm console, resulting in her sustaining serious injuries (MAIB report 10/2017¹¹).

While proceeding in parallel at a speed of around 6kts, the RIBs' skippers increased speed and then turned away from each other with the intention of completing synchronised power turns and returning to their parallel positions. As they exited the turns the skippers realised that the RIBs were in danger of colliding and reduced speed. Although they both acted quickly to reduce the speed of their respective vessels, and so lessen the impact, they were unable to prevent the collision. The manoeuvre had previously been carried out successfully on several occasions, but it had not been formally risk assessed, and no thought had been given to what to do if a collision situation developed.

¹¹ <https://www.gov.uk/maib-reports/collision-between-rigid-inflatable-boats-osprey-and-osprey-ii-resulting-in-serious-injuries-to-1-passenger>

SECTION 2 – ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 OVERVIEW

Rib Tickler and the PW collided because they were carrying out uncoordinated high-speed manoeuvres in the same stretch of water and neither the RIB driver nor the PW rider realised the risk until it was too late. Jane Walker died because she was struck by the PW and suffered internal injuries that were detailed by the pathologist as *essentially unsurvivable*.

This section of the report will analyse the circumstances that led to the collision and the reasons why the RIB and the PW were manoeuvring so close to each other at speed. The underlying factors that might have contributed to the accident, such as the skills, knowledge, and experience of those involved, local governance of the marine environment, available training, and the national oversight of PW will also be discussed.

2.3 THE COLLISION

Immediately before the collision, the PW rider jumped *Rib Tickler*'s wake at high speed. As the PW approached the wake, the RIB, which was being driven by the PW's owner, turned sharply to starboard and into the path of the PW. The PW hit *Rib Tickler*'s wake, rose out of the water, and struck the RIB owner's mother and the RIB simultaneously.

Within a few minutes of accepting the owner's offer to drive *Rib Tickler*, the RIB driver had accelerated the RIB to near full speed. Although the actual speed of the RIB could not be established, analysis of GoPro® footage recorded by the PW rider (**Figure 6**) showed the RIB's throttle handle at a position very close to full ahead. This would have resulted in the RIB travelling at between 25 and 30kts. Powerboat speeds of 25 to 30kts are not excessive and are common for RIBs operating in open and calm waters where there are no speed limits. However, given the confined nature of the waters, the proximity of the two PW, and the RIB driver's lack of powerboat training and experience driving a RIB, *Rib Tickler* was not being driven at a safe speed or in a safe manner.

The RYA *Personal Watercraft* handbook advised that PW should be driven at a speed that allows the rider to stay in control and take avoiding action if necessary. As with driving a car, when speed increases, the distance between vehicles also needs to increase. The safety label on the PW engine cover warned riders to keep a safe distance from people in the water and other watercraft and highlighted the potentially fatal consequences of a collision. It specifically warned riders not to follow directly behind other boats or jump over wakes or waves.

Neither the PW nor the RIB were being driven safely on the evening of the accident. However, despite the PW rider's lack of experience, there was little she could do to avoid the collision after committing to crossing the wake once *Rib Tickler* turned in to her path.

2.7.4 Oversight of inexperienced/untrained helms in an informal environment

There will be occasions when trained powerboat and PW owners wish to allow untrained or inexperienced friends and family to helm the craft, where local regulations and insurance policies allow.

In such a situation it would be prudent to give the inexperienced helm a short, informal familiarisation period, covering the basics of craft operation and safety. It would also be sensible for the owner to position themselves ready to intervene quickly if necessary. There is currently no advice from the RYA on how this should be achieved, and formalisation of what good practice looks like could enhance its course and guidance material.

2.7.5 Crossing waves and wakes

Encountering waves and wakes is inevitable when using PW and powerboats. The circumstances of these encounters differ hugely depending on, among other things, the environmental conditions, performance of the craft and proficiency of the helm. Scenarios range from being able to manage a craft in a significant sea state, to operating in a group of craft in sheltered waters, such as the Menai Strait, creating wakes for each other to jump.

Negotiating waves is covered in the RYA *Personal Watercraft* handbook, which includes a warning about the antisocial behavioural aspects of jumping the wake of other boats. However, there is currently no information on the dangers of wake jumping or lessons learned from previous accidents involving this hazardous activity. Sharing this information with the maritime leisure community would increase awareness and contribute to safer PW operation.

SECTION 3 – CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT

1. *Rib Tickler* and the PW collided because the RIB's helmsman turned into the PW's path as its rider jumped the RIB's wake. [2.2]
2. Jane Walker died because she was struck by the PW and suffered injuries that were *essentially unsurvivable*. [2.2]
3. *Rib Tickler* was not proceeding at a safe speed given the proximity of the PW, the amount of room available on the water and experience and knowledge of its driver. [2.3]
4. The PW rider did not leave sufficient distance between the PW and RIB to be able to take avoiding action at the speeds involved. [2.3]
5. The RIB driver did not have full awareness of other water users before commencing his turn to starboard. With little prior experience of helming a RIB, he did not have the necessary level of skill and knowledge to carry out manoeuvres at high speed within a group of craft. [2.4.1]
6. *Rib Tickler's* owner made an assumption about the ability of the RIB driver to take the helm. As a result, no attempt was made to provide any informal familiarisation or oversight. [2.4.2]
7. The PW rider did not have the necessary skills, knowledge, or experience to appreciate the hazards involved in wake jumping but was following the lead of the more experienced PW users within the family group. [2.4.3]
8. The two families had a misguided confidence in their abilities, which led to them not seeking the guidance and good practice offered by RYA courses and affiliation with local clubs and the safe practices they could offer. [2.4.4]

3.2 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. IACC did not have the resources necessary to manage its waterspace effectively. [2.5.1]
2. Authorities wishing to manage PW activity on their waters should seek advice and guidance from knowledgeable bodies such as PWP, MCA, UKHMA and BPA. [2.5.2]
3. IACC had not formally considered the impact of reducing the resources available to their maritime team, neither had risk assessments for foreseeable events been carried out. These factors and others, namely no clear incident reporting system, indicate the MSMS was in need of improvement. [2.5.3]
4. Though already taught in the RYA courses, greater emphasis could be placed within its handbooks and syllabi on the importance of over-the-shoulder pre-manoeuve checks for both powerboating and PW use. [2.7.2]

5. When removing the practice of pacing from their courses, the RYA removed some useful best practice for craft operating within a group, especially regarding speeds, distances, and communication. [2.7.3]
6. There is currently no advice from the RYA on providing informal familiarisation for or oversight of untrained helms, covering the basics of craft operation and safety. [2.7.4]
7. Negotiating waves is covered in the RYA *Personal Watercraft* handbook; however, there is currently no information on the dangers of wake jumping or lessons learned from previous accidents involving this hazardous activity. [2.7.5]

3.3 OTHER SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT

1. The legal status of PW since the 2005 Court of Appeal ruling made the regulation of irresponsible PW use difficult. It is hoped that the legislative changes sought by DfT will make management of these leisure activities easier to manage. [2.6.1]
2. The inconsistent approach to PW management around the UK coast has a detrimental impact on the efforts being made to reduce irresponsible PW use. The formalisation of a cross-industry forum with the ability, through its membership, to influence the majority of coastal and port authorities would facilitate the development of a nationally consistent approach. [2.6.2]